

Veeva Technical Assessment Last-Night Prep

Focus target: answer at least 20 MCQs correctly, solve SQL first, then medium coding, then attempt hard coding.

High-Yield MCQ Question Bank

Java Basics and OOP

1. Which keyword is used for inheritance in Java? - extends
2. Which keyword is used to implement an interface? - implements
3. What is polymorphism? - Ability of an object/method to take many forms
4. What is encapsulation? - Binding data and methods together and protecting data
5. Which access modifier supports encapsulation most? - private
6. What is abstraction? - Hiding internal details and showing essential features
7. What is inheritance? - Child class acquiring parent class properties
8. What is a class? - Blueprint for objects
9. What is an object? - Instance of a class
10. Which keyword creates an object? - new
11. Can constructor have return type? - No
12. Constructor name must be same as what? - Class name
13. When is constructor called? - When object is created
14. If no constructor is written, what does Java provide? - Default constructor
15. What does this keyword refer to? - Current object
16. What does static mean? - Belongs to class, not object
17. Are static variables shared by all objects? - Yes
18. Why is main method static? - JVM can call it without object creation
19. Method overloading means what? - Same method name, different parameters
20. Method overriding means what? - Child class redefines parent method
21. Can return type alone overload a method? - No
22. Is inheritance required for overriding? - Yes
23. Compile-time polymorphism is also called? - Method overloading
24. Runtime polymorphism is also called? - Method overriding
25. In `Animal a = new Dog(); a.sound();`, overridden method of which class runs? - Dog
26. Can Java support multiple inheritance using classes? - No
27. Can Java support multiple inheritance using interfaces? - Yes
28. Abstract class can have constructor? - Yes
29. Interface can have constructor? - No
30. Abstract class uses which keyword in child class? - extends
31. Interface uses which keyword in child class? - implements
32. Interface methods are by default what? - public and abstract
33. Can abstract class have normal methods? - Yes
34. Can abstract class have abstract methods? - Yes
35. What is final variable? - Value cannot be changed

- 36. What is final method? - Cannot be overridden
- 37. What is final class? - Cannot be inherited
- 38. What is super keyword used for? - Access parent class members/constructor
- 39. Which access modifier is accessible everywhere? - public
- 40. Which access modifier is accessible only inside same class? - private
- 41. Which modifier allows same package and subclasses? - protected
- 42. Default access is also called? - package-private
- 43. What is exception? - Runtime error/event disrupting normal flow
- 44. Which block always executes in exception handling? - finally
- 45. Which keywords are used in exception handling? - try, catch, finally, throw, throws
- 46. `10 / 0` causes which exception? - `ArithmeticException`
- 47. Accessing invalid array index causes what? - `ArrayIndexOutOfBoundsException`
- 48. Calling method on null causes what? - `NullPointerException`
- 49. What is garbage collection? - Reclaiming memory from unreachable objects
- 50. Which method suggests garbage collection? - `System.gc()`

Java Strings, Arrays, Operators, Loops

- 51. Are Java strings mutable or immutable? - Immutable
- 52. Which method gives string length? - `length()`
- 53. Which method gets character at index? - `charAt()`
- 54. Which method compares string content? - `equals()`
- 55. `==` compares what for objects? - References
- 56. `"5" + 5` gives what? - 55
- 57. `"Java".replace('J', 'P')` gives what? - Pava
- 58. Which class is best for mutable string operations? - `StringBuilder`
- 59. Array index starts from what? - 0
- 60. Last index of array is what? - `length - 1`
- 61. Accessing `arr[arr.length]` causes what? - `ArrayIndexOutOfBoundsException`
- 62. `int[] arr = {1, 2, 3}; arr.length` is what? - 3
- 63. Operator precedence: multiplication or addition first? - Multiplication
- 64. `10 + 5 * 20` equals what? - 110
- 65. `if (a == 2)` checks what? - Equality
- 66. `if (a = 2)` is valid for int in Java? - No
- 67. Loop `for (int i=0; i<=100; i+=5)` runs how many times? - 21
- 68. What does `continue` do? - Skips current iteration and moves to next
- 69. What does `break` do? - Exits loop/switch
- 70. `while (x > 0)` with x increasing from 1 is what? - Infinite loop
- 71. Enhanced for loop is used for what? - Iterating arrays/collections
- 72. Default value of int instance variable? - 0
- 73. Default value of boolean instance variable? - false
- 74. Default value of object reference? - null
- 75. Local variables have default values? - No

Collections, Generics, Threads

- 76. Root interface of collection hierarchy? - Collection
- 77. Which collection stores key-value pairs? - Map
- 78. Which class stores key-value pairs without sorted order? - HashMap
- 79. Which map maintains insertion order? - LinkedHashMap
- 80. Which map sorts by natural ordering of keys? - TreeMap
- 81. Which collection stores unique values? - Set
- 82. Which set usually has no order guarantee? - HashSet
- 83. Which collection allows duplicates and index access? - ArrayList
- 84. Which collection follows LIFO? - Stack
- 85. Which collection follows FIFO? - Queue
- 86. Stack insertion operation is called? - push
- 87. Stack deletion operation is called? - pop
- 88. Stack top read operation is called? - peek
- 89. Queue insertion operation is called? - enqueue/offer
- 90. Queue deletion operation is called? - dequeue/poll
- 91. Empty stack pop/peek causes what? - Invalid operation/exception
- 92. Can linked list implement stack? - Yes
- 93. Can linked list implement queue? - Yes
- 94. What is generic type erasure? - Generic type info removed at runtime
- 95. What is type inference? - Compiler infers generic type automatically
- 96. What is functional interface? - Interface with exactly one abstract method
- 97. Lambda captured local variables must be what? - final or effectively final
- 98. Which stream operation is terminal: collect or map? - collect
- 99. Purpose of Optional? - Handle possible null values safely
- 100. Structural modification during fail-fast iteration causes what? - ConcurrentModificationException
- 101. What is thread pool advantage? - Reuses existing threads and improves performance
- 102. Deadlock happens when? - Threads wait on locks held by each other
- 103. synchronized keyword is used for what? - Thread-safe access/locking
- 104. Local variables are stored where? - Stack memory
- 105. Objects are stored where? - Heap memory

DSA and Complexity

- 106. $O(1)$ means what? - Constant time
- 107. $O(n)$ means what? - Linear time
- 108. $O(n^2)$ usually comes from what? - Nested loops
- 109. $O(\log n)$ usually comes from what? - Halving/doubling each step
- 110. Binary search best case complexity? - $O(1)$
- 111. Binary search average/worst complexity? - $O(\log n)$
- 112. Sorting generally takes what time? - $O(n \log n)$
- 113. Two nested loops n and m give what? - $O(nm)$
- 114. Outer n with inner m and inner k sequentially gives what? - $O(nm + nk)$
- 115. Loop $i = i * 2$ until n gives what? - $O(\log n)$
- 116. Loop $j * j \leq n$ gives what? - $O(\sqrt{n})$
- 117. First block $n \sqrt{m}$, second block $m \sqrt{n}$ gives what? - $O(n\sqrt{m} + m\sqrt{n})$

- 118. Huffman tree building complexity? - $O(n \log n)$
- 119. Linear search complexity? - $O(n)$
- 120. HashMap average search complexity? - $O(1)$
- 121. HashSet average contains complexity? - $O(1)$
- 122. Stack push/pop complexity? - $O(1)$
- 123. Queue enqueue/dequeue complexity? - $O(1)$
- 124. Array random access complexity? - $O(1)$
- 125. Linked list random access complexity? - $O(n)$
- 126. Linked list insertion at head complexity? - $O(1)$
- 127. Maximum subarray problem uses which algorithm? - Kadane's algorithm
- 128. Kadane's algorithm complexity? - $O(n)$
- 129. Two sum optimized uses what? - HashMap/HashSet
- 130. Two sum optimized complexity? - $O(n)$
- 131. Pair sum in sorted array uses what? - Two pointers
- 132. Palindrome check uses what? - Two pointers
- 133. Longest substring without repeating uses what? - Sliding window + HashSet
- 134. Fixed-size subarray max sum uses what? - Sliding window
- 135. Move zeroes to end uses what? - Two pointer/index pointer
- 136. Rotate array optimized uses what? - Reverse method
- 137. Count frequency uses what? - HashMap or frequency array
- 138. Anagram lowercase optimized uses what? - Frequency array
- 139. General anagram uses what? - HashMap
- 140. First non-repeating character uses what? - Frequency map/array
- 141. Remove duplicates uses what? - HashSet
- 142. Reverse string uses what? - Two pointers/StringBuilder
- 143. Check subsequence uses what? - Two pointers
- 144. Merge sorted arrays uses what? - Two pointers
- 145. Greedy means what? - Choose locally best option at each step
- 146. DP means what? - Store answers to overlapping subproblems
- 147. Climbing stairs recurrence? - $dp[i] = dp[i-1] + dp[i-2]$
- 148. 0/1 knapsack uses which technique? - Dynamic Programming
- 149. First Fit Decreasing bin packing first step? - Sort items descending
- 150. Group consecutive cards uses what? - Sort + HashMap frequency

DBMS and SQL MCQs

- 151. DBMS stands for what? - Database Management System
- 152. Table stores data in what? - Rows and columns
- 153. Row is also called what? - Record/tuple
- 154. Column is also called what? - Attribute/field
- 155. Primary key property? - Unique and not null
- 156. Foreign key is used for what? - Refers to primary key of another table
- 157. Candidate key means what? - Possible unique identifier
- 158. Super key means what? - Any set of attributes uniquely identifying row
- 159. Unique key can allow null? - Yes, depending on DBMS

- 160. Primary key can allow null? - No
- 161. Normalization reduces what? - Redundancy/duplication
- 162. 1NF removes what? - Repeating groups/non-atomic values
- 163. 2NF removes what? - Partial dependency
- 164. 3NF removes what? - Transitive dependency
- 165. ACID full form? - Atomicity, Consistency, Isolation, Durability
- 166. Atomicity means what? - All or nothing
- 167. Consistency means what? - Valid state to valid state
- 168. Isolation means what? - Transactions do not interfere
- 169. Durability means what? - Committed data remains saved
- 170. DDL commands? - CREATE, ALTER, DROP, TRUNCATE
- 171. DML commands? - INSERT, UPDATE, DELETE
- 172. DQL command? - SELECT
- 173. TCL commands? - COMMIT, ROLLBACK, SAVEPOINT
- 174. DCL commands? - GRANT, REVOKE
- 175. WHERE filters what? - Rows before grouping
- 176. HAVING filters what? - Groups after GROUP BY
- 177. ORDER BY does what? - Sorts result
- 178. GROUP BY does what? - Groups rows by column values
- 179. JOIN does what? - Combines rows from tables
- 180. INNER JOIN returns what? - Matching rows only
- 181. LEFT JOIN returns what? - All left rows plus matching right rows
- 182. RIGHT JOIN returns what? - All right rows plus matching left rows
- 183. FULL OUTER JOIN returns what? - All matching and non-matching rows from both tables
- 184. CROSS JOIN returns what? - Cartesian product
- 185. MySQL supports FULL OUTER JOIN directly? - No
- 186. Subquery means what? - Query inside another query
- 187. IN operator checks what? - Value exists in a list/subquery result
- 188. EXISTS checks what? - Whether subquery returns rows
- 189. DISTINCT removes what? - Duplicate rows
- 190. MAX returns what? - Maximum value
- 191. AVG returns what? - Average value
- 192. COUNT counts what? - Rows/non-null values depending usage
- 193. EXCEPT does what? - Returns rows from first query not in second
- 194. UNION does what? - Combines results and removes duplicates
- 195. UNION ALL does what? - Combines results keeping duplicates
- 196. Window function for ranking transactions? - ROW_NUMBER()
- 197. PARTITION BY does what in window functions? - Divides rows into groups
- 198. ORDER BY inside ROW_NUMBER does what? - Decides numbering order
- 199. CASE WHEN in SQL is used for what? - Conditional logic
- 200. ROUND(AVG(x),1) returns what? - Average rounded to 1 decimal

Mock-Pattern Direct Questions

- 201. Which keyword is used to inherit a class? - extends

- 202. Which answer best describes polymorphism? - Object can take many forms
- 203. Which is not valid overloading? - Changing only return type
- 204. Which is true about abstract classes and interfaces? - Abstract classes can have constructors, interfaces cannot
- 205. Which modifier allows same package and subclass access? - protected
- 206. `int d = 10 + 5 * 20` output? - 110
- 207. Most likely deadlock scenario? - Thread holding one lock and waiting for another held by another thread
- 208. Primary advantage of thread pool? - Reusing existing threads
- 209. Java feature omitting generic method type arguments? - Type inference
- 210. Generic type information at runtime? - Completely erased
- 211. How are TreeMap elements ordered? - Natural ordering of keys
- 212. Binary search best case? - $O(1)$
- 213. Outer n with inner m and k loops complexity? - $O(nm + nk)$
- 214. Linked list efficiently implements what? - Stack and Queue
- 215. Huffman tree build complexity? - $O(n \log n)$
- 216. Loss exactly $1/5$ of revenue means what equation? - Expenditure - Revenue = Revenue / 5
- 217. Online orders but never store orders uses which set operation? - Online EXCEPT Store
- 218. Committed transaction saved after failure is which ACID property? - Durability
- 219. Join without specific matching column condition? - CROSS JOIN
- 220. First Fit Decreasing starts by sorting how? - Descending order
- 221. Consecutive card grouping requires divisibility by what? - `hand.length % groupSize == 0`
- 222. SQL customer category filter can use what clause after grouping? - HAVING
- 223. Third transaction per user uses what function? - ROW_NUMBER()
- 224. `ROW_NUMBER() OVER(PARTITION BY user_id ORDER BY transaction_date)` gives what? - Chronological transaction rank per user
- 225. Average cook/pack/delay per item uses what? - GROUP BY Item with AVG

Must-Practice Coding Problems

- 1. Count frequency of array elements - HashMap
- 2. Find duplicates in array - HashSet
- 3. Two Sum - HashMap
- 4. Pair sum in sorted array - Two pointers
- 5. Reverse array/string - Two pointers
- 6. Move zeroes to end - Two pointers
- 7. Rotate array by k - Reverse method
- 8. Second largest element - One pass
- 9. Maximum subarray sum - Kadane's algorithm
- 10. Palindrome string - Two pointers
- 11. Anagram - Frequency array/HashMap
- 12. First non-repeating character - Frequency map
- 13. Remove duplicate characters - HashSet
- 14. Longest substring without repeating - Sliding window + HashSet
- 15. Maximum sum subarray of size k - Fixed sliding window
- 16. Minimum size subarray sum - Variable sliding window

17. Valid parentheses - Stack
18. Next greater element - Stack
19. Queue operation tracing - Queue
20. Reverse linked list - Iterative pointers
21. Merge two sorted linked lists - Two pointers
22. Climbing stairs - DP
23. Coin change basics - DP
24. First Fit Decreasing bin packing - Greedy + sorting
25. Group consecutive cards - Sorting + HashMap

Must-Know SQL Query Templates

1. Employee with department name

```
SELECT e.name, d.dept_name
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id;
```

2. All employees even without department

```
SELECT e.name, d.dept_name
FROM employees e
LEFT JOIN departments d ON e.dept_id = d.dept_id;
```

3. Salary above average

```
SELECT name, salary
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

4. Second highest salary

```
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

5. Count employees department-wise including zero

```
SELECT d.dept_name, COUNT(e.emp_id) AS employee_count
FROM departments d
LEFT JOIN employees e ON d.dept_id = e.dept_id
GROUP BY d.dept_name;
```

6. Customers who bought Electronics but not Clothing/Footwear

```
SELECT c.customer_id, c.name
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
JOIN Order_Items i ON o.order_id = i.order_id
JOIN Products p ON i.product_id = p.product_id
GROUP BY c.customer_id, c.name
HAVING SUM(CASE WHEN p.category = 'Electronics' THEN 1 ELSE 0 END) > 0
AND SUM(CASE WHEN p.category IN ('Clothing', 'Footwear') THEN 1 ELSE 0 END) = 0;
```

7. Average values per item

```
SELECT Item,
       ROUND(AVG(Cook_time), 1) AS Average_cook,
       ROUND(AVG(Pack_time), 1) AS Average_pack,
       ROUND(AVG(Delay_time), 1) AS Average_delay
FROM Orders
GROUP BY Item;
```

8. Third transaction per user

```
WITH ranktran AS (  
    SELECT user_id, spend, transaction_date,  
           ROW_NUMBER() OVER (  
               PARTITION BY user_id  
               ORDER BY transaction_date ASC  
           ) AS rn  
    FROM transactions  
)  
SELECT user_id, spend, transaction_date  
FROM ranktran  
WHERE rn = 3;
```

9. Online-only customers using EXCEPT

```
SELECT customer_id  
FROM orders  
WHERE order_type = 'Online'  
EXCEPT  
SELECT customer_id  
FROM orders  
WHERE order_type = 'Store';
```

10. Highest paid job titles

```
SELECT DISTINCT worker_title  
FROM title  
WHERE worker_ref_id IN (  
    SELECT worker_id  
    FROM worker  
    WHERE salary = (SELECT MAX(salary) FROM worker)  
)  
ORDER BY worker_title;
```

Final Night Priority

1. Memorize MCQs 1-225.
2. Revise SQL templates 1-10.
3. Code only these: Two Sum, longest substring, valid parentheses, Kadane, rotate array, move zeroes, group consecutive cards.
4. Sleep enough. Tomorrow morning, revise formulas and templates only.